Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it has decayed and time t_1 when $\frac{1}{3}$ of it had decayed is:

(A) $14\ \text{min}$

(B) $20\ \text{min}$

(C) $28\ \text{min}$

(D) $7\ \text{min}$

Q.2

At a certain instant, a radioactive sample has a decay rate of 5000 disintegrations per minute. After 5 minutes, the decay rate is 1250 disintegrations per minute. The decay constant (per minute) is

(A) $0.4\ln(2)$

(B) $0.2\ln(2)$

(C) $0.1\ln(2)$

(D) $0.8\ln(2)$

Q.3

The activity of a radioactive sample is measured as N_0 counts per minute at $t = 0$ and N_0/e counts per minute at $t = 5$ minutes. The time (in minutes) at which the activity reduces to half its value is

(A) $\log_e 2/5$

(B) $\frac{5}{\log_e 2}$

(C) $5\log_{10} 2$

(D) $5\log_e 2$

Q.4

Beta rays emitted by a radioactive material are

(A) electromagnetic radiations

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(B) the electrons orbiting around the nucleus

(C) charged particles emitted by the nucleus

(D) neutral particles

Q.5

Two radioactive materials X_1 and X_2 have decay constants 10λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $1/e$ after a time

(A) $\frac{1}{10\lambda}$

(B) $\frac{1}{11\lambda}$

(C) $\frac{11}{10\lambda}$

(D) $\frac{1}{9\lambda}$

Q.6

The half life of a radioactive isotope ' X' is 50 years. It decays to another element 'Y' which is stable. The two elements ' X' and ' Y ' were found to be in the ratio of $1:15$ in a sample of a given rock. The age of the rock was estimated to be

(A) 150 years

(B) 200 years

(C) 250 years

(D) 100 years

Q.7

Let N_β be the number of β particles emitted by gram of Na^{24} radioactive nuclei (half life $=15$ hrs) in 7.5 hours, N_β is close to (Avogadro number $=6.023 \times 10^{23} / \mathrm{g}$ mole) :

(A) 6.2×10^{21}

(B) 7.5×10^{21}

(C) 1.25×10^{22}

(D) 1.75×10^{22}

Q.8

The activity of a radioactive sample is A_1 at time and A_2 at time t_1. If τ is average life of sample then the number of nuclei decayed in time $(t_2 - t_1)$ is

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(A) $A_1 t_1 - A_2 t_2$

(B) $\frac{(A_1 - A_2)}{2}\tau$

(C) $(A_1 - A_2)(t_2 - t_1)$

(D) $(A_1 - A_2)\tau$.

Q.9

Two radioactive substances A and B have decay constants 5λ and λ respectively. At $\mathrm{t} = 0$ they have the same number of nuclei. The ratio of number of nuclei of A to those of B will be $(1/e)^2$ after a time interval

(A) 4λ

(B) 2λ

(C) $1/2\lambda$

(D) $1/4\lambda$

Q.10

When ${}_3\mathrm{L}^7$ nuclei are bombared by protons and the resultant nuclei are ${}_4\mathrm{Be}^8$, the emitted particles will be

(A) alpha particle

(B) beta particles

(C) gamma photons

(D) nentrons

Q.11

Which of the following is in the increasing order for penetrating power

(A) α, β, γ

(B) β, α, γ

(C) γ, α, β

(D) γ, β, α

Q.12

Consider a radioactive material of half-life 1.0 minute. If one of the nuclei decays now, the next one will decay

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(A) After 1 minute

(B) After $\frac{1}{\log_e 2}$ minute

(C) After $\frac{1}{N}$ minute, where N is the number of nuclei present at that moment

(D) After any time

Q.13

Which of the following is a correct statement?

(A) Beta rays are same as cathode rays

(B) Gamma rays are high energy neutrons

(C) Alpha particles are singly ionized helium atoms

(D) Protons and neutrons have exactly the same mass

Q.14

The half-life of radioactive Radon is 3.8 days. The time at the end of which $1/20^{\text{th}}$ of the Radon sample will remain undecayed is
(Given $\log_{10} \mathrm{e} = 0.4343$)

(A) 3.8 days

(B) 16.5 days

(C) 33 days

(D) 76 days

Q.15

A gamma ray creates an electron-positron pair. If the rest mass energy of an electron is $0.5\mathrm{MeV}$ and the total kinetic energy of the electron positron pair is $0.78\mathrm{MeV}$, then the energy of the gamma ray must be

(A) $0.78\mathrm{MeV}$

(B) $1.78\mathrm{MeV}$

(C) $1.28\mathrm{MeV}$

(D) $0.28\mathrm{MeV}$

Q.16

In the nuclear fusion reaction

$$ {}_1^2\mathrm{H} + {}_1^3\mathrm{H} \rightarrow {}_2^4\mathrm{He} + n $$

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given that the repulsive potential energy between the two nuclei is $\sim 7.7 \times 10^{-14}$ J, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's Constant $k = 1.38 \times 10^{-23}$ J/K]

(A) 10^7 K

(B) 10^5 K

(C) 10^3 K

(D) 10^9 K

Q.17

When Uranium is bombarded with neutrons, it undergoes fission. The fission reaction can be written as : ${}_{92}\mathrm{U}^{235} + {}_0n^1 \rightarrow {}_{56}\mathrm{Ba}^{141} + {}_{36}\mathrm{Kr}^{92} + 3x + \mathrm{Q}$
where three particles named x are produced and energy Q is released. What is the name of the particle x?

(A) electron

(B) α -particle

(C) neutron

(D) neutrino

Q.18

Determine the power output of $\mathrm{a}_{92}\mathrm{U}^{235}$ reactor if it takes 30 days to use 2 kg of fuel. Energy released per fission is $200\mathrm{MeV}$ and $N = 6.023 \times 10^{26}$ per kilomole.

(A) $63.28\mathrm{MW}$

(B) $3.28\mathrm{MW}$

(C) $0.6\mathrm{MW}$

(D) $50.12\mathrm{MW}$

Q.19

For fission of uranium ${}^{235}_{92}\mathrm{U}$, one needs

(A) very fast neutrons (a few electrons volts)

(B) low energy neutrons $(\approx 0.04\mathrm{eV}$ at 300 K$)$

(C) a moderator making neutrons slow

(D) both (b) and (c)

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Q.20

A 1.5×10^9 years old rock contains ^{238}U which disintegrates to form ^{206}Pb. Assume that there was no lead in the rock initially and it is the only stable product formed by the decay. Calculate the ratio of the number of nuclei of lead to that of uranium. [Given, half-life of $^{238}U = 4.5 \times 10^9$ years, $2^{\frac{1}{3}} \approx 1.26$]

(A) 0.16

(B) 0.36

(C) 0.46

(D) 0.26

Answers & Solutions

Q.1 Answer:

20 min

Solution:

$N = N_0 e^{-\lambda t}$

At time $t = t_1, N = \frac{N_0}{3}$**. Therefore**

$\frac{N_0}{3} = N_0 e^{-\lambda t} \Rightarrow t_1 = \frac{1}{\lambda}(\ln 3)$

At time $t = t_2$, $N = \frac{2N_0}{3}$**. Therefore**

$$\frac{2N_0}{3} = N_0^{-\lambda t_2} \Rightarrow t_2 = \frac{1}{\lambda}\left(\ln \frac{3}{2}\right)$$

$$\therefore \quad t_1 - t_2 = \frac{1}{\lambda}\left(\ln 3 - \ln \frac{3}{2}\right)$$

$$= \frac{1}{\lambda}(\ln 3 - \ln 3 + \ln 2) = \frac{1}{\lambda} \ln 2$$

If $T_{1/2}$ **is the half life,**

$T_{1/2} = \frac{\ln 2}{\lambda}$

$\therefore t_2 - t_1 = T_{1/2} = 20$ min

Q.2 Answer:

$0.4 \ln(2)$

Solution:

Decay rate $|R| = \lambda N$

Let N_0 **be the number of radioactive nuclei present at an instant when** $|R| = 5000$ **disintegrations per minute. Then**

$5000 = \lambda N_0$ **(1)**

After $t = 5$ **min, we have**

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$$1250 = \lambda N$$

$$\Rightarrow \lambda = \frac{1250}{N} \qquad (2)$$

Using

$$5000 = \frac{1250 N_0}{N}$$

$$\Rightarrow \quad N = \frac{N_0}{4}$$

Now $N = N_0 e^{-\lambda t} = N_0 e^{-5\lambda}$

$\Rightarrow \quad \frac{N}{N_0} = e^{-5\lambda} \Rightarrow \frac{1}{4} = e^{-5\lambda}$

or $4 = e^{5\lambda} \Rightarrow \ln(4) = 5\lambda$

$\Rightarrow \quad 2\ln(2) = 5\lambda$

$\Rightarrow \quad \lambda = \frac{2\ln(2)}{5} = 0.4\ln(2)$

Q.3 Answer:

$5 \log_e 2$

Solution:

$N = N_0 e^{-\lambda t}$

Here, $t = 5$ **minutes**

$\frac{N_0}{e} = N_0 \cdot e^{-5} \Rightarrow \quad 5\lambda = 1, \lambda = \frac{1}{5},$

Now, $T_{1/2} = \frac{\ell n 2}{\lambda} = 5\ell n 2$

Q.4 Answer:

charged particles emitted by the nucleus

Solution:

β **-particles are charged particles emitted by the nucleus.**

Q.5 Answer:

$\frac{1}{9\lambda}$

Solution:

$N_1 = N_0 e^{-10\lambda t}$ **and** $N_2 = N_0 e^{-\lambda t}$.

$\therefore \quad \frac{N_1}{N_2} = \frac{e^{-10\lambda t}}{e^{-\lambda t}} = \frac{1}{e^{9\lambda t}}$

Given $\frac{N_1}{N_2} = \frac{1}{e}; \quad \therefore \quad \frac{1}{e^{9\lambda t}} = \frac{1}{e}$

or, $9\lambda t = 1$ **or** $t = \left(\frac{1}{9\lambda}\right)$

Q.6 Answer:

200 years

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Solution:

Let number of atoms in $\mathrm{X}=\mathrm{N}_{\mathrm{x}}$ Number of atoms in $\mathrm{Y}=\mathrm{N}_{\mathrm{y}}$
By question $\frac{\mathrm{N}_{\mathrm{x}}}{\mathrm{N}_{\mathrm{y}}}=\frac{1}{15}$
$\therefore$ Part of $\mathrm{N}_{\mathrm{x}}=\frac{1}{16}\left(\mathrm{N}_{\mathrm{x}}+\mathrm{N}_{\mathrm{y}}\right)=\frac{1}{2^4}\left(\mathrm{N}_{\mathrm{x}}+\mathrm{N}_{\mathrm{y}}\right)$
So, total 4 half lives are passed, so, age of rock is $4 \times 50=200$ years

Q.7 Answer:

7.5×10^{21}

Solution:

We know that $N_\beta=N_0\left(1-e^{-\lambda t}\right)$
$N_\beta=\frac{6.023 \times 10^{23}}{24}\left[1-e^{\frac{\ell \mathrm{n} 2}{15}} \times 7.5\right]$
on solving we get, $\mathrm{N}_\beta=7.4 \times 10^{21}$
If α and B are emitted simultaneously.

Q.8 Answer:

$\left(A_1-A_2\right) \tau$.

Solution:

Let N_0 be the initial number of nuclei, then $N_1=N_0 e^{-\lambda t_1}$ and $N_2=N_0 e^{-\lambda t_2}$
$\therefore \quad$ Number of nuclei decayed $=N_1-N_2$
$=N_0\left(e^{-\lambda t_1}-e^{-\lambda t_2}\right)=\frac{A_0}{\lambda}\left(e^{-\lambda t_1}-e^{-\lambda t_2}\right)$
$=\frac{A_1-A_2}{\lambda}=\left(A_1-A_2\right) \tau$

Q.9 Answer:

$1/2\lambda$

Solution:

$\lambda_A=5 \lambda$ and $\lambda_B=\lambda$
At $t=0,\left(N_0\right)_A=\left(N_0\right)_B$
Given, $\frac{N_A}{N_B}=\left(\frac{1}{e}\right)^2$
According to radioactive decay, $\frac{N}{N_0}=e^{-\lambda t}$
$\therefore \frac{N_A}{\left(N_0\right)_A}=e^{-\lambda_A t}$ and $\frac{N_B}{\left(N_0\right)_B}=e^{-\lambda_B t}$
From (1) and (2), $\frac{N_A}{N_B}=e^{-(5 \lambda-\lambda) t}$
$\Rightarrow\left(\frac{1}{e}\right)^2=e^{-4 \lambda t}=\left(\frac{1}{e}\right)^{4 \lambda t} \Rightarrow 4 \lambda t=2 \therefore t=\frac{1}{2 \lambda}$

Q.10 Answer:

gamma photons

Solution:

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$_{P}^{1} + {}_{3}\mathrm{Li}^{7} = {}_{4}\mathrm{Be}^{\mathrm{B}} + \gamma$ **(energy)**

Q.11 Answer:

α, β, γ

Solution:

Penetration power of γ is 100 times of B, while that of B is 100 times of α,

Q.12 Answer:

After any time

Solution:

Because radioactivity is a spontaneous phenomenon.

Q.13 Answer:

Beta rays are same as cathode rays

Solution:

Both the β -rays and the cathode rays are made up of electrons but γ -rays are EM waves, α -particles
are doubly ionized helium atoms and protons and neutrons have approximately the same mass.

Q.14 Answer:

16.5 days

Solution:

By the formula $\mathbf{N = N_0 e^{-\lambda t}}$
Given $\frac{\mathrm{N}}{\mathrm{N}_0} = \frac{1}{20}$ **and** $\lambda = \frac{0.6931}{3.8} \Rightarrow 20 = \mathrm{e}^{\frac{0.6931 \times t}{3.8}}$
Taking log of both sides
or $\log 20 = \frac{0.6931 \times t}{3.8} \log_{10} \mathrm{e}$
or $1.3010 = \frac{0.6931 \times t \times 0.4343}{3.8} \Rightarrow t = 16.5$ **days.**

Q.15 Answer:

$1.78\mathrm{MeV}$

Solution:

Energy of γ -ray photon = Rest mass energy +
K.E. $= 2(0.5) + 0.78 = 1.78\mathrm{MeV}$

Q.16 Answer:

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10^9 K

Solution:

average kinetic energy per molecule $= \frac{3}{2}kT$

This kinetic energy should be able to provide the repulsive potential energy

$\therefore \frac{3}{2}kT = 7.7 \times 10^{-14}$

$\Rightarrow T = \frac{2 \times 7.7 \times 10^{-14}}{3 \times 1.38 \times 10^{-23}} = 3.7 \times 10^9$ K

Q.17 Answer:

neutron

Solution:

No solution available

Q.18 Answer:

63.28MW

Solution:

Number of atoms in 2 kg fuel $= \frac{2}{235} \times 6.023 \times 10^{26} = 5.12 \times 10^{24}$

number of fission per second $= \frac{5.12 \times 10^{24}}{30 \times 24 \times 60 \times 60} = 1.978 \times 10^{18}$

Energy released per fission $= 200\text{MeV} = 200 \times 1.6 \times 10^{-13} = 3.2 \times 10^{-11}$ J

Power output $= 3.2 \times 10^{-11} \times 1.978 \times 10^{18}$

$= 63.28 \times 10^6$ W $= 63.28$MW

Q.19 Answer:

both (b) and (c)

Solution:

Fission of $^{235}_{92}\text{U}$ **is possible by only slow neutrons. To produce slow neutrons, one has to use moderators so that fission is possible.**

Q.20 Answer:

0.26

Solution:

Let N_0 **be the initial number of uranium nuclei. After time** t**, let** N_U **be the number of uranium nuclei and** N_Pb **be the number of lead nuclei.**

The number of half-lives passed is

$n = \frac{t}{T_\text{half}} = \frac{1.5 \times 10^9}{4.5 \times 10^9} = \frac{1}{3}$

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$N_{\mathrm{U}} = N_0 \frac{1}{2}^{n} = N_0 \frac{1}{2}^{\frac{1}{3}}$

$N_{\mathrm{Pb}} = N_0 - N_0 \frac{1}{2}^{\frac{1}{3}}$

$\frac{N_{\mathrm{Pb}}}{N_{\mathrm{U}}} = 2^{\frac{1}{3}} - 1 = 0.26$

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